

Inhibition of Marine Biofouling by Use of Degradable and Hydrolyzable Silyl Acrylate Copolymer

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S Supporting Information

ABSTRACT: Silyl acrylate copolymers are promising materials for marine antibiofouling. However, their structures need optimizing to improve their erosion and mechanical properties. We have prepared copolymer of 2-methylene-1,3-dioxepane (MDO), tributylsilyl methacrylate (TBSM) and methyl methacrylate (MMA) via radical ring-opening copolymerization. Such polymer has a degradable backbone and hydrolyzable side groups. Our study demonstrates that as the ester units in the backbone increase, the degradation rate increases but the swelling decreases in seawater. The degradation is controlled by the polymer composition or the molar ratio of the ester units in the backbone to the silyl ester side groups. Moreover, such polymer can serve as a carrier and controlled release system for organic antifoulants. Marine field tests show that the system consisting of the copolymer and organic antifoulant has good antifouling performance depending on the polymer composition. It can effectively inhibit the colonization and growth of marine organisms when MDO content is above 20 wt %.

1. INTRODUCTION

Marine biofouling occurring on submerged surfaces such as vessels, heat exchangers, oceanographic sensors and aquaculture facilities brings a number of problems to marine industries.^{1,2} To prevent and control biofouling, coatings containing various biocides such as copper, arsenic, mercury and tin oxide were employed.³ Particularly, tributyltin based self-polishing coating had dominated the market until they were prohibited in 2008 due to its serious ecological impact.^{3–5} Today, it is urgent to develop environment-friendly systems for marine antibiofouling.

Marine antibiofouling deals with fabrication of antifouling materials and development of efficient antifoulants.^{6,7} As regards the former, various materials such as poly(dimethylsiloxane) elastomers,^{8–10} PEG copolymers,^{11,12} zwitterionic polymers,^{13,14} and self-generated hydrogel,^{15,16} slippery liquid-infused surfaces¹⁷ have been prepared. However, their long-term antifouling ability in marine environments needs testing. On the other hand, some green antifoulants such as 4,5-dichloro-2-octyl-isothiazolone, 4(5)-[1-(2,3-dimethylphenyl)-ethyl]-1H-imidazole and butenolide have been developed.^{18,19} Their long-term antifouling performance in field is under evaluation. Actually, the controlled release of the antifoulants is critical for their performance, which mainly depends on the erosion of the polymer carrying them.²⁰

Self-polishing copolymers (SPCs) which generally refer to the acrylic copolymers with hydrolyzable pendant groups, for example, silyl acrylate, zinc or copper acrylate polymer, have been used in marine antibiofouling coatings since the hydrolysis of the pendant groups and the dissolution of the resulting hydrolyzed-polymer can lead to a self-polishing surface and the release of incorporated biocides.^{21,22} Trialkylsilyl-based (meth)acrylate copolymer is the most environment-friendly in SPCs. It also exhibits stable hydrolysis rate and self-smoothing.⁵

However, the self-polishing ability of such a polymer under static conditions, depending on the content of silyl ester side groups and the seawater motion, is limited because it has a nonreactive main chain.^{5,23} It is important to optimize its structure to improve its properties. Bressy et al. developed diblock silyl ester copolymer with *tert*-butyldimethylsilyl ester as side groups by RAFT polymerization. The block polymer shows more stable erosion rate and better antifouling ability than random silyl ester copolymer.^{24–26} They also improved the thermal stability and erosion property of poly(*tert*-butyldimethylsilyl methacrylate) by copolymerization with poly(dimethylsiloxane) macromonomers.^{27,28} Generally, a high silyl ester content (normally above 50 wt %) is required for SPCs to achieve a good self-polishing ability.¹⁶ Yet, the polymers with high silyl ester content are so hydrophilic that they are swollen after immersion in seawater, which declines their adhesion to substrate and antibiofouling ability.²⁹ Thus, SPCs with good self-polishing property but small swelling are desired.

Recently, we have developed marine antibiofouling materials based on polyurethane with poly(ϵ -caprolactone) (PCL) segments in the main chain and poly(triisopropylsilyl acrylate) (PTIPSA) side chains.²³ The system consisting of the polyurethane and antifoulants exhibits good antifouling performance. However, preparation procedure for the polyurethane is tedious. 2-Methylene-1,3-dioxepane (MDO) can copolymerize with vinyl monomers via radical ring-opening polymerization (RROP), where ester units are incorporated into backbone to generate a new class of biodegradable

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Scheme 1. Synthesis of Poly(MDO-co-TBSM-co-MMA) (PMSMx) via RROP

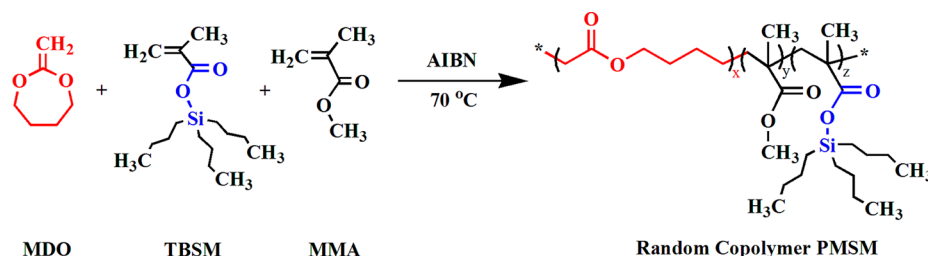


Table 1. Characterization Data of PMSMx

sample	copolymer composition (wt %) ^a			$M_{n, GPC} \times 10^{-4}$ (g/mol) ^b	$\bar{M}_w$ ^b	yield ^c (wt %)	T_g (°C) ^d	$T_{50\%}$ (°C) ^e
	MDO	TBSM	MMA					
PMSM0	0	48	52	17.4	2.25	94	54.4	339.4
PMSM8	8	41	51	14.3	4.62	70	36.6	391.4
PMSM23	23	23	54	8.9	4.15	66	2.0	394.0
PMSM38	38	14	48	7.3	4.14	60	−23.0	383.6
PMSM47	47	0	53	6.3	4.21	58	−19.7	398.8

^aDetermined by ^1H NMR. ^bDetermined by GPC. ^cCalculated by weight. ^dDetermined by DSC. ^eTemperature at 50% weight loss determined by TGA.

materials (poly(vinyl-co-ester)s).^{30–34} It provides a novel and facile approach to chemically modify silyl acrylate polymers. In the present study, we have prepared degradable silyl acrylate copolymers consisting of 2-methylene-1,3-dioxepane (MDO), tributylsilyl methacrylate (TBSM) and methyl methacrylate (MMA) by RROP. The polymer has degradable and hydrophobic backbone as well as hydrolyzable side groups. Such a structure improves its self-polishing but decreases its swelling. The antifouling performance and mechanical properties of the polymer can be modulated by varying its composition. Moreover, the polymer degrades or erodes without the help of shear force, which facilitates the antibiofouling in static conditions. We have investigated the self-polishing, swelling, antifoulant release and marine antibiofouling of the polymer in marine environment as a function of the polymer composition. We attempt to develop high performance marine antibiofouling systems.

2. EXPERIMENTAL SECTION

Materials. 2-Methylene-1,3-dioxepane (MDO) and tributylsilyl methacrylate (TBSM) were synthesized by using procedures detailed elsewhere.^{35–38} Methyl methacrylate (MMA) (Aladdin) was purified by distillation before use. Azobis(isobutyronitrile) (AIBN) (Aladdin) was recrystallized three times from methanol. 4,5-Dichloro-2-octyl-isothiazolone (DCOIT) was kindly provided by Thankful Chemical (Guangzhou). Artificial seawater (ASW) was prepared following ASTM D1141-98 (2013). Other reagents were used as received.

Synthesis of MDO, TBSM, and MMA Copolymer (PMSMx). As described in Scheme 1, PMSMx was synthesized by RROP in the bulk. A certain amount of MDO, TBSM, MMA, and AIBN (1 mol % of the total monomers) were added into a Schlenk tube, and the reaction mixture was degassed by three vacuum/nitrogen cycles. The polymerization was performed under nitrogen at 70 °C for 3 h. The resulting polymer was dissolved in chloroform, precipitated in methanol, and then dried for 48 h. ^1H NMR (600 MHz, CDCl_3 , ppm): 4.01 (–CH₂COOCH₂–), 3.63 (–OCH₃), 2.28

(–CH₂COOCH₂–), 1.42–1.74 (–CH₂COOCH₂CH₂CH₂–), 1.32 (–SiCH₂CH₂CH₂CH₃), 0.88 (–SiCH₂CH₂CH₂CH₃), 0.75 (–SiCH₂CH₂CH₂CH₃). The copolymer is designed as PMSMx, where x is the weight percentage of MDO determined by ^1H NMR. The characterization data of PMSMx are shown in Table 1.

Characterization. Proton Nuclear Magnetic Resonance Spectroscopy (^1H NMR). ^1H NMR spectra were recorded on a Bruker AV600 NMR spectrometer using CDCl_3 as the solvent and tetramethylsilane as the internal standard.

Gel Permeation Chromatography (GPC). The number-average molecular weight and the polydispersity were determined by a Waters 1515 GPC equipped with a Waters 2414 refractive index detector. The columns used were of Styragel HR 2, HR 4, and HR 6. Tetrahydrofuran (THF) was used as the eluent at the flow rate of 1.0 mL/min. Calibrations were performed using polystyrene standards (1.30×10^3 g/mol to 2.21×10^6 g/mol).

Differential Scanning Calorimetry (DSC). DSC measurement was performed on a NETZSCH DSC 204F1 differential scanning calorimeter under a nitrogen flow of 50 mL/min. After the sample was quickly heated to 130 °C and equilibrated at the temperature for 5 min to remove thermal history, it was cooled to −80 °C at a rate of 10 °C/min. Then, it was heated to 130 °C at a rate of 10 °C/min. The glass transition temperature (T_g) was obtained from the endothermic shift during the second heating scan. Only one T_g can be observed, indicating that the resulting polymer is random copolymer. Moreover, PMSMx has a lower T_g than PMSM0, and the T_g decrease with MDO content (Table 1). This is because the incorporation of MDO units makes the polymer chain more flexible. Note that molecular weight should slightly affect T_g in the range we investigated (6×10^4 to 17×10^4 g/mol).³⁹

Thermogravimetric Analysis (TGA). TGA measurement was performed on a NETZSCH TG 209F1 instrument under nitrogen atmosphere at a heating rate of 10 °C/min in the range from 30 to 800 °C. PMSMx samples have higher decomposition temperature ($T_{50\%}$, 380–400 °C) than PMSM0 (~340 °C) (Table 1), indicating that the introduction of MDO

in the main chain can improve the thermal stability of silyl acrylate copolymer.

Pull-Off Test. The adhesion of PMSMx film with the epoxy resin primer was measured using Pull-Off Adhesion Tester (PosiTest AT-A Automatic) on an aluminum dolly (20 mm diameter) (ASTM D 4541-89). PMSMx film with a thickness of $\sim 300\ \mu\text{m}$ was prepared via solution casting where the polymer concentration in xylene was 30% (w/v). Five different points on each sample were tested to obtain an average value.

Hydrolytic Degradation and Swelling. The test was conducted in ASW at $25\ ^\circ\text{C}$. The PMSMx film on an epoxy resin panel ($20 \times 20\ \text{mm}^2$ in dimension) was prepared via a solution-casting. Generally, a solution of 30% (w/v) polymers in xylene was dripped onto the panel and kept at room temperature for 1 week to remove the residual solvent. The weight (W_0) of each dried coating together with its panel was measured before the coating was immersed in ASW that was replaced every 2 weeks. At certain intervals, the sample was taken out, dipped in Milli-Q water for 5 min, then gently dried with filter paper before its weight ($W_{t,\text{wet}}$) was recorded. In parallel, the sample was dried at $25\ ^\circ\text{C}$ for 1 day and then at $60\ ^\circ\text{C}$ before their weights ($W_{t,\text{dry}}$) were recorded. The mass loss (wt %) and the water adsorption (wt %) were calculated by the following equation:

$$\text{mass loss (wt \%)} = \frac{W_0 - W_{t,\text{dry}}}{W_0 - W_{\text{panel}}} \times 100\% \quad (1)$$

$$\text{water adsorption (wt \%)} = \frac{W_{t,\text{wet}} - W_{t,\text{dry}}}{W_{t,\text{dry}} - W_{\text{panel}}} \times 100\% \quad (2)$$

where W_0 , $W_{t,\text{dry}}$, $W_{t,\text{wet}}$, and W_{panel} are the initial weight of the panel coated with PMSMx, the wet weight of the coated panel at time t , the dry weight of the coated panel at time t , and the weight of the panel without coating, respectively. For each sample, three coated panels were prepared and measured.

Contact Angle (CA) Measurement and Attenuated Total Reflectance Fourier Transform Infrared Spectroscopy (ATR-FTIR). The stationary CA measurement was conducted on a Contact Angle System OCA40 (Dataphysics) at $25\ ^\circ\text{C}$ by using the sessile method. Five different points on each sample were tested to obtain an average value. FTIR spectra were recorded using a Bruker Vertex 70 FTIR spectrometer in ATR mode in the range of $600\ \text{cm}^{-1}$ to $4000\ \text{cm}^{-1}$ at room temperature with a resolution of $4\ \text{cm}^{-1}$ and an accumulation of 32 scans. The samples were obtained from ASW at different time intervals, immersed in Milli-Q water for 5 min, and dried at $25\ ^\circ\text{C}$ before the CA and ATR-FTIR measurements.

Surface Roughness Measurement. The surface roughness measurement was performed using a BMT Expert 3D surface profile measurement system. PMSMx film was prepared via a solution-casting with a thickness about $250\ \mu\text{m}$. For each sample, six lines in the range of 3 mm were scanned to obtain an average value of surface roughness (μm). The surface topography was observed by an area-scan measurement with a dimension of $0.5 \times 0.5\ \text{mm}^2$.

Antifoulant Release Rate Measurement. The release rate of DCOIT was determined during a period of immersion in ASW at $25\ ^\circ\text{C}$ according to ASTM D6903–07 (2013). The sample solution is prepared in the proportion of 90 wt % polymers with 10 wt % DCOIT in xylene with a solid content of 30 wt %, and then it was coated onto a designed cylinder with an area about $100\ \text{cm}^2$. The coated cylinder was dried at room

temperature for a week and then immersed in ASW in a holding tank. ASW was replaced every 2 days in case of the saturation of DCOIT in ASW. After certain time, the cylinder was taken out and fixed on a rotating instrument which controlled the cylinders rotating at $60 \pm 5\ \text{rpm}$ for 2 h in a container loaded with 1.5 L of ASW. Afterward, DCOIT was extracted by 10 mL hexane from 100 mL ASW and its concentration in the hexane was determined quantitatively by an Agilent Cary 60 UV–vis spectrophotometer.

Marine Field Tests. The field tests were performed from November 2014 to March 2015 in South China Sea ($22^\circ 33'\text{N}$, $114^\circ 31'\text{E}$) in China, where the temperature of surface water ranges from 25 to $28\ ^\circ\text{C}$ and pH (~ 8.2) as well as salinity ($\sim 35\ \text{‰}$) are relatively stable. The samples were applied onto the glass fiber reinforced epoxy resin panels ($100 \times 100 \times 3\ \text{mm}^3$) and then lowered into seawater at a depth of $\sim 1\ \text{m}$. After a certain period of time, the panels were taken out of the sea and carefully washed with seawater and photographed, and then they were immediately placed into the seawater to continue the test. Glass fiber reinforced epoxy resin panels as a control sample and panels coated with PMSMx were tested.

3. RESULTS AND DISCUSSION

We first examined the adhesion strength of PMSMx to substrate, which is critical for applications of marine antibiofouling material. Figure 1 shows the pull off strength

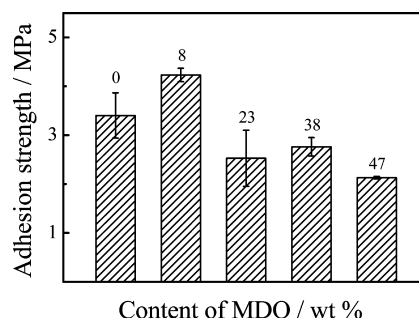


Figure 1. Adhesion strength of PMSMx with different MDO content.

of the polymer to epoxy panel. For PMSM0, the adhesion strength is about 3.4 MPa. When MDO is introduced, the adhesion strength varies. For PMSM8, PMSM23, PMSM38, and PMSM47, the adhesion strength values are 4.2, 2.5, 2.8, and 2.1 MPa, respectively. Note that PMSM8 has the highest adhesion strength. PMSM8 has a more flexible chain than PMSM0, so it has more contacts with the substrate. That is why PMSM8 has higher adhesion strength than PMSM0. As MDO content increases above 23 wt %, the adhesion strength decreases and it is even lower than PMSM0. This is because the contacts between the polymer and substrate decrease as tributyl silyl groups decrease. Anyhow, each PMSMx has adhesion strength above 2 MPa, which is enough for the antifouling coating.⁴⁰

Figure 2 shows the time dependence of mass loss of PMSMx after immersion in ASW due to the degradation of ester groups on main chain and hydrolysis of silyl ester side groups. As MDO content increases from 8 wt % to 23 wt %, the mass loss increases. This is due to the scission of the ester linkages on the backbone which makes the resulting polymer dissolve into seawater more easily. However, the mass loss decreases when MDO content is higher than 23 wt %. This is because TBSM

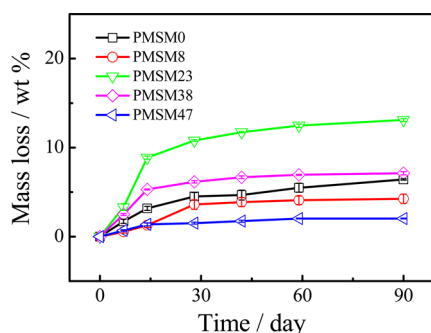


Figure 2. Time dependence of mass loss of PMSMx after immersing in ASW at 25 °C.

content decreases as MDO content increases, so the anionic carboxylate groups decrease and the hydrolyzation of the polymer becomes more difficult. Accordingly, the erosion of the polymer depends on the combination of the degradation of its backbone and the hydrolysis of tributylsilyl side groups. Compared with PMSM0 and PMSM47, PMSM23, and PMSM35 show higher erosion rate due to the cooperative effect between their degradation and the solubility. PMSM0 without MDO in the main chain has a mass loss higher than PMSM8, and it is even higher than that of PMSM47 without silyl ester groups. As discussed above, the degradation of polymer backbone and the hydrolysis of tributylsilyl side groups contribute to the erosion, but their contributions are not equal. The former is less than the latter. We can control the erosion by varying the polymer composition or the molar ratio of silyl ester side groups to ester units in main chain. Note that temperature, pH, and salinity of the marine environment can also influence the degradation. Here, they are relatively stable, their effects can be neglected.

Figure 2 also shows that each film has quick mass loss in the first 14 days. This is probably caused by a quick release of some low molar mass compounds such as residual solvents.⁴¹ Actually, the antifoulants are stably released in that the mass loss is still going in 90 days. It is well established that the hydrolysis and degradation of hydrophobic self-polishing copolymers generally happen on polymer surface contacting with seawater.^{3–5} As the polymer degrades, the degraded fragments leave the surface and dissolve in seawater, so the surface is continuously polished, leading to the release of the antifoulants.^{21,22} We estimated the apparent thickness and degradation rate as a function of time (Figure S1). Except that the thickness shows a significant decrease and the degradation rate has an increase in the initial stage (within 14 days) due to the release of the low molar mass compounds, generally, the thickness decreases while the degradation rate is almost constant, further indicating the degradation happens on the polymer surface. In other words, the thickness slightly influences the degradation and fouling control. However, the thickness is critical for the service life of self-polishing polymer coatings. As the thickness increases, the duration increases.

Figure 3 shows the water absorption rates of PMSMx after immersion in ASW. As MDO content increases or TBSM content decreases, the water adsorption of polymer decreases from 78% to 3%. As discussed above, the tributylsilyl ester groups can be hydrolyzed into anionic carboxylate groups. More tributylsilyl ester groups in PMSMx yield more anionic carboxylate groups, leading to higher water absorption. In other words, the incorporation of MDO into silyl acrylate copolymer

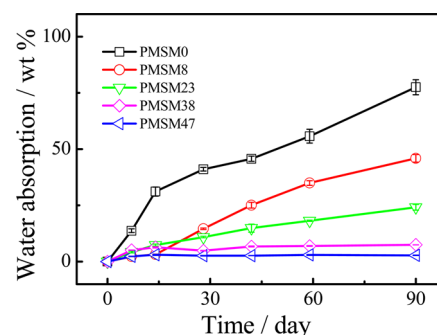


Figure 3. Time dependence of swelling of PMSMx film in ASW at 25 °C.

can effectively reduce the swelling of polymer. Note that T_g of the polymer decreases with MDO content. However, the decrease in T_g does not increase the water adsorption since the hydrophobicity increases as MDO content increases. Thus, the water adsorption is determined by tributylsilyl ester groups.

Figure 4 shows contact angle of PMSMx after immersion in ASW. Before immersion, PMSMx surface has a CA above 90°,

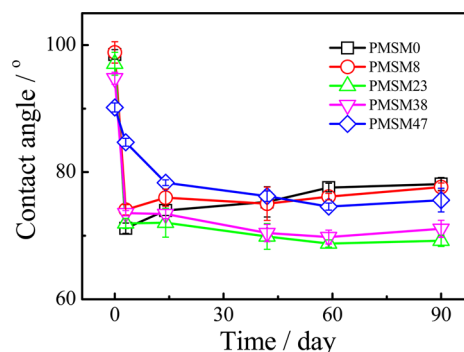


Figure 4. Time dependence of contact angle of PMSMx after immersing in ASW at 25 °C.

indicating it is hydrophobic. After immersion for 3 days, the CA decreases to 75°, indicating PMSMx containing TBSM units becomes more hydrophilic due to the presence of anionic carboxylate groups after the hydrolysis of tributylsilyl ester groups. Fourteen days later, the CA levels off, indicating that the polymers are stably hydrophilic in seawater.

Figure 5 shows the typical ATR-FTIR spectra of the PMSMx before and after immersion in ASW. After 3 days, the new band at 1554 cm^{-1} assigned to the asymmetrical stretching vibration of $-\text{COONa}$ appears in PMSM0 or PMSM23, indicating the hydrolysis of tributylsilyl ester group. The bands are still there even after 90 days, indicating the surface composition is stable. For PMSM47, we cannot observe such a band throughout the immersion. This is understandable because there are no hydrolyzable silyl ester groups in PMSM47.

Figure 6 shows the time dependence of surface roughness of PMSMx after immersing in ASW at 25 °C. After one month, the surface roughness increases from 0.06 to 0.34 μm to 0.14–0.49 μm , indicating the occurrence of hydrolysis. After two months, the surface roughness tends to level off, indicating that PMSMx exhibits self-smoothing ability in ASW with constant erosion rate. Clearly, PMSM23 has the lowest surface roughness. As discussed above, PMSM23 has the fastest erosion with small swelling. Namely, it has good self-polishing

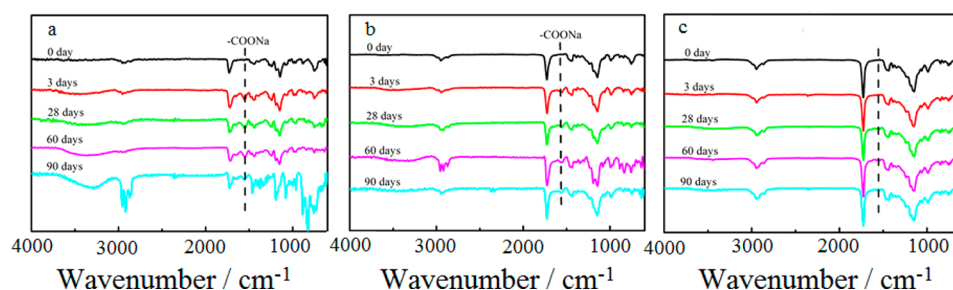


Figure 5. ATR-FTIR spectra of PMSM0 (a), PMSM23 (b), and PMSM47 (c) before and after immersing in ASW at 25 °C.

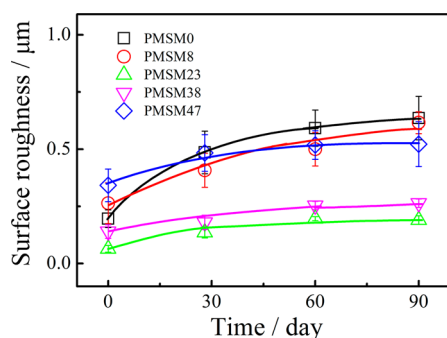


Figure 6. Time dependence of surface roughness of PMSMx in ASW at 25 °C.

ability. Its low surface roughness is favorable for reducing drag.⁴²

Figure 7 shows the typical DCOIT release rates measured from the test cylinders rotated at a speed of 60 ± 5 rpm. The

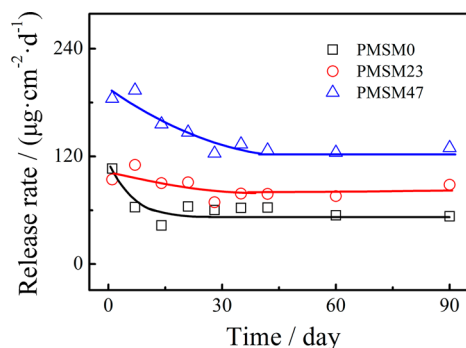
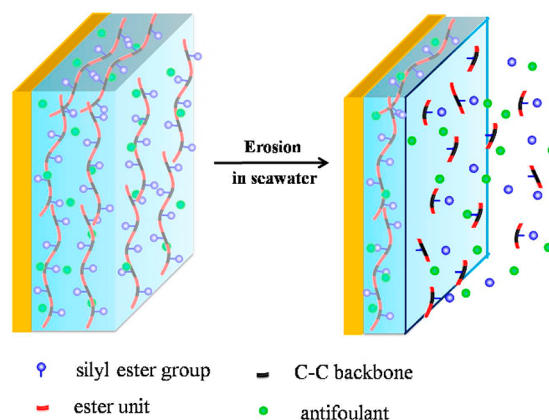


Figure 7. Time dependence of DCOIT release rate from PMSMx at 25 °C.

initial release rate for PMSM0, PMSM23, and PMSM47 are $94 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$, $106 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$ and $185 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$, respectively. After immersion in ASW for 30 days, the release rates are stable at $55 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$, $80 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$ and $120 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$, respectively. As discussed above, the self-polishing of PMSM23 is faster than that of PMSM0, so PMSM23 has a higher release rate. Clearly, the introduction of MDO can improve the release of DCOIT. Note that PMSM47 has the highest release rate even though its erosion rate is the lowest. The possible reason is that the copolymer chain would degrade into more fragments as MDO content increases, which is favorable to the release of the antifoulants. Anyhow, the release rate of antifoulant increases with MDO content. The mechanism about the release of antifoulants from the polymer

is illustrated in Scheme 2. With the degradation of the polymer, the surface gradually erodes and the antifoulants are released.

Scheme 2. Mechanism about the Release of Antifoulants from the Polymer



The antifouling at molecular level of materials is generally monitored by protein release. However, such a method is not applicable to the system consisting of self-polishing polymer and biocide because the protein release and antifoulant release happen at the same time and they are difficult to distinguish. In the present study, we evaluated the antifouling performance of the self-polishing copolymer with DCOIT by using cyprids of barnacle *B. Amphitrite* as the fouling model in lab assay. All the cyprids were dead on the polymer coating surface, indicating the system can effectively inhibit the growth of marine fouling. Particularly, we evaluated the long-term antifouling performance of the system by marine field tests. Figure 8 shows the marine field test results for PMSMx-based coatings in South China Sea and the quantitative analysis of the biofouling. After one month, the control panel with epoxy surfaces is already seriously fouled by organisms, indicating a high fouling pressure. However, for PMSMx coatings containing DCOIT, the surfaces are clean at early stage of immersion. The surface of PMSM0 or PMSM8 becomes uneven due to swelling (Figure 2). After 4 months, the surfaces of PMSM0 and PMSM8 are seriously swollen and fouled by marine organisms. In contrast, the surfaces of PMSMx with MDO more than 20 wt % MDO are still clean, indicating that they have good antibiofouling ability. Clearly, the copolymer with enough MDO can release enough DCOIT to inhibit the biofouling. On the other hand, the swelling significantly decreases with MDO content, indicating that the incorporation of MDO can improve the mechanical properties in marine environment.

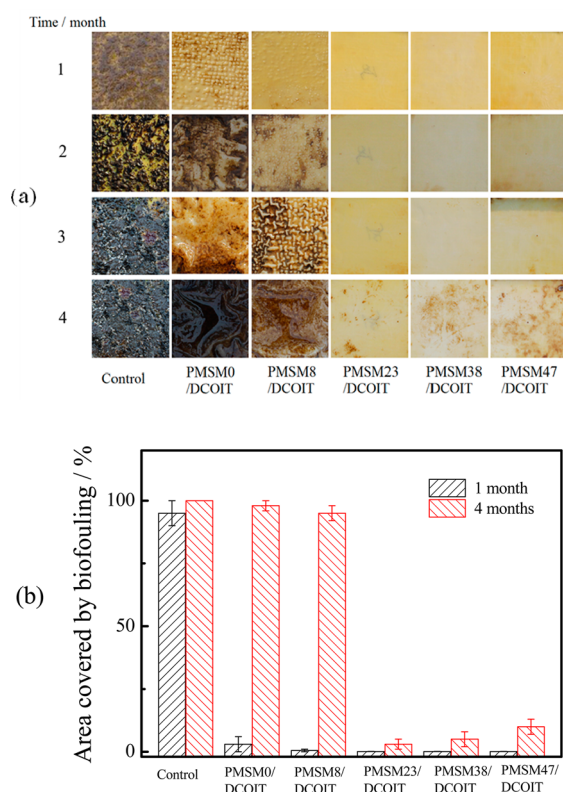


Figure 8. (a) Typical images of tested panels coated with PMSMx with DCOIT (10 wt %) after immersion in seawater. (b) The quantitative analysis of biofouling estimated from the area covered by foulants.

4. CONCLUSION

We have prepared chemically modified silyl acrylate copolymers by radical ring-opening copolymerization. Compared with the traditional silyl acrylate copolymers, the silyl acrylate copolymer with polyester segments in the backbone has higher degradation rate but lower swelling. The degradation and antifouling performance of the polymer can be controlled by varying the polymer composition. When MDO content is above 20 wt %, modified silyl acrylate copolymer based coating exhibits excellent antibiofouling ability in marine environment for 4 months. We believe the degradable silyl acrylate copolymer is promising in development of long-term marine antibiofouling coatings.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.5b01819.

Figure S1(PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Buskens, P.; Wouters, M.; Rentrop, C.; Vroon, Z. A brief review of environmentally benign antifouling and foul-release coatings for marine applications. *J. Coat. Technol. Res.* **2013**, *10*, 29–36.
- (2) Banerjee, I.; Pangule, R. C.; Kane, R. S. Antifouling coatings: recent developments in the design of surfaces that prevent fouling by proteins, bacteria, and marine organisms. *Adv. Mater.* **2011**, *23*, 690–718.
- (3) Almeida, E.; Diamantino, T. C.; de Sousa, O. Marine paints: The particular case of antifouling paints. *Prog. Org. Coat.* **2007**, *59*, 2–20.
- (4) Chambers, L. D.; Stokes, K. R.; Walsh, F. C.; Wood, R. J. K. Modern approaches to marine antifouling coatings. *Surf. Coat. Technol.* **2006**, *201*, 3642–3652.
- (5) Bressy, C.; Margailan, A.; Fay, F.; Linossier, I.; Réhel, K. Tin-free self-polishing marine antifouling coatings. In *Advances in Marine Antifouling Coatings and Technologies*; Hellio, C.; Yebra, D. M. Y., Eds.; Woodhead Publishing, 2009; pp 445–491.
- (6) Callow, J. A.; Callow, M. E. Trends in the development of environmentally friendly fouling-resistant marine coatings. *Nat. Commun.* **2011**, *2*, 244.
- (7) Gittens, J. E.; Smith, T. J.; Suleiman, R.; Akid, R. Current and emerging environmentally-friendly systems for fouling control in the marine environment. *Biotechnol. Adv.* **2013**, *31*, 1738–1753.
- (8) Fang, J.; Kellarakis, A.; Wang, D.; Giannelis, E. P.; Finlay, J. A.; Callow, M. E.; Callow, J. A. Fouling release nanostructured coatings based on PDMS-polyurea segmented copolymers. *Polymer* **2010**, *51*, 2636–2642.
- (9) Vucko, M. J.; Poole, A. J.; Carl, C.; Sexton, B. A.; Glenn, F. L.; Whalan, S.; de Nys, R. Using textured PDMS to prevent settlement and enhance release of marine fouling organisms. *Biofouling* **2014**, *30*, 1–16.
- (10) Fernández Estarlich, F. M.; Eaton, P. J.; Fletcher, R. L.; Lewey, S. A.; Nevell, T. G.; Smith, J. R.; Tsibouklis, J. The Effects of Incorporated Silicone Oils and Calcium Carbonate on the Resistance to Settlement and the Antifouling Performance of a Silicone Elastomer. *J. Adhes. Sci. Technol.* **2011**, *25*, 2183–2198.
- (11) Zhou, Z.; Calabrese, D. R.; Taylor, W.; Finlay, J. A.; Callow, M. E.; Callow, J. A.; Fischer, D.; Kramer, E. J.; Ober, C. K. Amphiphilic triblock copolymers with PEGylated hydrocarbon structures as environmentally friendly marine antifouling and fouling-release coatings. *Biofouling* **2014**, *30*, 589–604.
- (12) Gudipati, C. S.; Finlay, J. A.; Callow, J. A.; Callow, M. E.; Wooley, K. L. The Antifouling and Fouling-Release Performance of Hyperbranched Fluoropolymer (HBFP)-Poly(ethylene glycol) (PEG) Composite Coatings Evaluated by Adsorption of Biomacromolecules and the Green Fouling Alga *Ulva*. *Langmuir* **2005**, *21*, 3044–3053.
- (13) Quintana, R.; Janczewski, D.; Vasantha, V. A.; Jana, S.; Lee, S. S.; Parra-Velandia, F. J.; Guo, S.; Parthiban, A.; Teo, S. L.; Vancso, G. J. Sulfobetaine-based polymer brushes in marine environment: is there an effect of the polymerizable group on the antifouling performance? *Colloids Surf., B* **2014**, *120*, 118–124.
- (14) Zhang, Z.; Finlay, J. A.; Wang, L.; Gao, Y.; Callow, J. A.; Callow, M. E.; Jiang, S. Polysulfobetaine-Grafted Surfaces as Environmentally Benign Ultralow Fouling Marine Coatings. *Langmuir* **2009**, *25*, 13516–13521.
- (15) Xie, L. Y.; Hong, F.; He, C. X.; Ma, C. F.; Liu, J. H.; Zhang, G. Z.; Wu, C. Coatings with a self-generating hydrogel surface for antifouling. *Polymer* **2011**, *52*, 3738–3744.
- (16) Hong, F.; Xie, L. Y.; He, C. X.; Liu, J. H.; Zhang, G. Z.; Wu, C. Effects of hydrolyzable comonomer and cross-linking on anti-biofouling terpolymer coatings. *Polymer* **2013**, *54*, 2966–2972.
- (17) Xiao, L. L.; Li, J. S.; Mieszkun, S.; Di Fino, A.; Clare, A. S.; Callow, M. E.; Callow, J. A.; Grunze, M.; Rosenhahn, A.; Levkin, P. A. Slippery Liquid-Infused Porous Surfaces Showing Marine Antibiofouling Properties. *ACS Appl. Mater. Interfaces* **2013**, *5*, 10074–10080.

- (18) Qian, P. Y.; Xu, Y.; Fusetani, N. Natural products as antifouling compounds: recent progress and future perspectives. *Biofouling* **2009**, *26*, 223–234.
- (19) Jacobson, A. H.; Willingham, G. L. Sea-nine antifoulant: an environmentally acceptable alternative to organotin antifoulants. *Sci. Total Environ.* **2000**, *258*, 103–110.
- (20) Carteau, D.; Vallée-Réhel, K.; Linossier, I.; Quiniou, F.; Davy, R.; Compère, C.; Delbury, M.; Fay, F. Development of environmentally friendly antifouling paints using biodegradable polymer and lower toxic substances. *Prog. Org. Coat.* **2014**, *77*, 485–493.
- (21) Kiil, S.; Weinell, C. E.; Pedersen, M. S.; Dam-Johansen, K. Analysis of self-polishing antifouling paints using rotary experiments and mathematical modeling. *Ind. Eng. Chem. Res.* **2001**, *40*, 3906–3920.
- (22) Kiil, S.; Dam-Johansen, K.; Weinell, C. E.; Pedersen, M. S.; Codolar, S. A. Dynamic simulations of a self-polishing antifouling paint exposed to seawater. *J. Coat. Technol.* **2002**, *74*, 89–91.
- (23) Xu, W. T.; Ma, C. F.; Ma, J. L.; Gan, T. S.; Zhang, G. Z. Marine biofouling resistance of polyurethane with biodegradation and hydrolyzation. *ACS Appl. Mater. Interfaces* **2014**, *6*, 4017–4024.
- (24) Bressy, C.; NGuyen, M. N.; Tanguy, B.; Ngo, V. G.; Margaillan, A. Poly(trialkylsilyl methacrylate)s: A family of hydrolysable polymers with tuneable erosion profiles. *Polym. Degrad. Stab.* **2010**, *95*, 1260–1268.
- (25) Bressy, C.; Hellio, C.; Nguyen, M. N.; Tanguy, B.; Maréchal, J. P.; Margaillan, A. Optimized silyl ester diblock methacrylic copolymers: A new class of binders for chemically active antifouling coatings. *Prog. Org. Coat.* **2014**, *77*, 665–673.
- (26) Nguyen, M. N.; Bressy, C.; Margaillan, A. Synthesis of novel random and block copolymers of *tert*-butyldimethylsilyl methacrylate and methyl methacrylate by RAFT polymerization. *Polymer* **2009**, *50*, 3086–3094.
- (27) Lejars, M.; Margaillan, A.; Bressy, C. Well-defined graft copolymers of *tert*-butyldimethylsilyl methacrylate and poly-(dimethylsiloxane) macromonomers synthesized by RAFT polymerization. *Polym. Chem.* **2013**, *4*, 3282.
- (28) Duong, T. H.; Bressy, C.; Margaillan, A. Well-defined diblock copolymers of poly(*tert*-butyldimethylsilyl methacrylate) and poly-(dimethylsiloxane) synthesized by RAFT polymerization. *Polymer* **2014**, *55*, 39–47.
- (29) Naoya, N.; Yasuto, H.; Yasuyuki, K.; Makoto, T. Antifouling coating composition, coating film formed from said antifouling coating composition, antifouling method using said antifouling coating composition and hull or underwater structure coated with said coating film. U.S. Patent. 6,110,990, 2000.
- (30) Undin, J.; Finne-Wistrand, A.; Albertsson, A. C. Adjustable degradation properties and biocompatibility of amorphous and functional poly(ester-acrylate)-based materials. *Biomacromolecules* **2014**, *15*, 2800–2807.
- (31) Cai, T.; Chen, Y.; Wang, Y.; Wang, H.; Liu, X.; Jin, Q.; Agarwal, S.; Ji, J. Functional 2-methylene-1,3-dioxepane terpolymer: a versatile platform to construct biodegradable polymeric prodrugs for intracellular drug delivery. *Polym. Chem.* **2014**, *5*, 4061–4068.
- (32) Sun, L. F.; Zhou, R. X.; Liu, Z. L. Synthesis and enzymatic degradation of 2-methylene-1,3-dioxepane and methyl acrylate copolymers. *J. Polym. Sci., Part A: Polym. Chem.* **2003**, *41*, 2898–2904.
- (33) Endo, T.; Yokozawa, T., Free radical ring-opening polymerization. In *New Methods for Polymer Synthesis*, Mijs, W. J., Ed.; Springer, 1992; pp 155–177.
- (34) Agarwal, S. Chemistry, chances and limitations of the radical ring-opening polymerization of cyclic ketene acetals for the synthesis of degradable polyesters. *Polym. Chem.* **2010**, *1*, 953.
- (35) Bailey, W. J.; Ni, Z.; Wu, S. R. Synthesis of poly(ϵ -caprolactone) via a free radical mechanism. Free radical ring-opening polymerization of 2-methylene-1,3-dioxepane. *J. Polym. Sci., Polym. Chem. Ed.* **1982**, *20*, 3021–3030.
- (36) Plikk, P.; Tyson, T.; Finne-Wistrand, A.; Albertsson, A. C. Mapping the characteristics of the radical ring-opening polymerization of a cyclic ketene acetal towards the creation of a functionalized polyester. *J. Polym. Sci., Part A: Polym. Chem.* **2009**, *47*, 4587–4601.
- (37) Mancardi, D.; Sindt, M.; Paul, J. M.; Mieloszynski, J. L. Convenient synthesis of tributylsilyl methacrylate. *Synth. Commun.* **2007**, *37*, 3873–3878.
- (38) Liu, G. B. Preparation of silyl esters by ZnCl_2 -catalyzed dehydrogenative cross-coupling of carboxylic acids and silanes. *Synlett* **2006**, *2006*, 1431–1433.
- (39) Plazek, D. J.; Ngai, K. L. *The Glass Temperature in Physical Properties of Polymers Handbook*; Mark, J. E., Ed.; Springer, 2007, p 192.
- (40) Rath, S. K.; Chavan, J. G.; Sasane, S.; Jagannath; Patri, M.; Samui, A. B.; Chakraborty, B. C. Two component silicone modified epoxy foul release coatings: Effect of modulus, surface energy and surface restructuring on pseudobarnacle and macrofouling behavior. *Appl. Surf. Sci.* **2010**, *256*, 2440–2446.
- (41) Yonehara, Y.; Yamashita, H.; Kawamura, C.; Itoh, K. A new antifouling paint based on a zinc acrylate copolymer. *Prog. Org. Coat.* **2001**, *42*, 150–158.
- (42) Schultz, M. P. Effects of coating roughness and biofouling on ship resistance and powering. *Biofouling* **2007**, *23*, 331–341.